ABF-YOLO: A Fine-Grained PCB Defect Detection Framework Integrating Axial Attention and Bidirectional Feature Fusion

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***Abstract*—Printed circuit board (PCB) defect detection is essential for ensuring manufacturing reliability and quality. However, existing methods often struggle with background noise and exhibit limited effectiveness in identifying small-scale defects. This study introduces a novel framework, denoted as ABF-YOLO, to overcome these limitations. This framework enhances PCB defect detection by integrating axial attention and bidirectional feature fusion (BiFusion). In this architecture, the axial attention module is introduced to refine fine-level feature representations and increase precision in detecting minor defects. The BiFusion module is embedded to facilitate multi-scale information fusion, enhancing the adaptability of the model to varying object dimensions. Experiments on a benchmark PCB defect dataset demonstrate that ABF-YOLO attains 97.67% mAP@0.5, 95.78% precision, and 95.83% recall, outperforming YOLOv12 by margins of +8.16%, +0.56%, and +13.49%, respectively. ABF-YOLO exhibits outstanding detection precision and robustness for small-sized defects even under complex background conditions, highlighting its practical effectiveness for challenging PCB inspection scenarios.**

***Index Terms*—Printed circuit board (PCB), defect detection, attention mechanism, feature fusion, YOLO.**

I. Introduction

PRINTED circuit boards (PCBs) serve as a core infrastructure in electronics, delivering mechanical stability and enabling electrical pathways between components. Even minor defects in PCBs may severely affect device functionality, potentially resulting in failures and major financial damage [1]. Critical failure modes, such as short and open circuits, are particularly harmful. Short circuit triggers conductive burnout, while open circuit disrupts signal paths, both of which have been identified as primary failure modes in field-returned high-density PCB assemblies [2]. Furthermore, the estimated costs of PCB failure significantly vary across industries, where consumer electronics mainly suffer from economic losses (more th an $10 billion) and product recalls, while aerospace and medical devices face critical safety concerns [3]. Recent investigations have reported that PCB failures can inflict large financial costs and even safety risks, emphasizing the need for highly sensitive automated detection of defects [4]. Despite significant advancements in manufacturing technology, the complete removal of PCB defects remains complex, necessitating robust detection strategies to address this pressing engineering issue in electronic production.

The advent of automated techniques leveraging computer vision and image-based analysis has led to a moderate enhancement in PCB defect detection efficiency [5], [6]. However, a number of these approaches still face limitations regarding practical industrial applicability. For instance, electrical detection methods [7], which locate conductive defects through electrical signal analysis, can effectively detect conductive defects but fail to identify non-conductive ones and may even cause secondary damage. Thermal imaging methods [8], which detect defects inside multilayer PCBs using thermal radiation properties, have limited practical application due to high equipment costs and insufficient accuracy in detecting surface micro-defects. Although automatic X-ray inspection [9] can penetrate inter-layer structures of PCBs to detect defects and produce high-quality imaging, its inspection speed is relatively slow and poses radiation safety risks. In addition, low-dose X-ray setups have been developed to minimize radiation exposure, which requires prolonged exposure times and resolution compromise, with limited performance of fine surface defect detection [10]. A hybrid inspection system that integrates multiple modalities leverages the strength of each modality and provides a well-balanced solution in terms of coverage, speed, and safety [11]. However, the practical implementation of a hybrid detection system remains constrained by certain challenges, such as data synchronization, increased hardware cost, and reduced throughput efficiency. Although these methods have partially improved the efficiency of PCB defect detection, significant technical challenges remain, particularly the low detection accuracy of micro-defects in practical applications.

The rapid evolution of deep learning has accelerated the growth and broadened the application scope of PCB defect detection technologies. Convolutional neural networks (CNNs), a representative class of deep learning models, have exhibited notable effectiveness in defect identification and are widely applied in industrial settings. Algorithms, such as Faster R-CNN [12] and YOLO [13], have shown impressive results in industrial image analysis and defect

Received 8 May 2025; revised 2 July 2025 and 29 July 2025; accepted 6 August 2025. Date of publication 8 August 2025; date of current version 7 November 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62373186, and in part by the Natural Science Foundation of Jiangsu Province under Grant BK20231440. *(Corresponding author: Mingyang Xie.)*

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Digital Object Identifier 10.1109/TCE.2025.3597022

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localization due to their precision and real-time processing capability. Liu et al. [14] recently introduced YOLO-pdd, a multi-scale PCB inspection approach that strengthens feature representation by combining deep sequential features and layered residual structures, significantly improving the detection of small-scale defects. Law et al. [15] developed an ensemble-based framework that incorporates detectors, such as MobileNet SSDv2 [16], Faster R-CNN [12], and YOLOv5 [17], which collectively enhance detection accuracy and generalization across diverse defect categories. One-stage and two-stage detectors, such as EfficientDet [18] and Faster R-CNN with FPN [19], have also been widely used in defect inspection due to the advantages of real-time detection and accuracy. However, these traditional CNN-based approaches often struggle with ultra-small defects (sub-100 μm cracks or micro-vias). The ultra-small defects exhibit low visual contrast, blurred boundaries, and minimal pixel footprint, making them vulnerable to being lost during convolution and down sampling processes, thereby posing a significant challenge for standard CNN feature extractors [20]. Despite substantial gains from recent deep-learning approaches in industrial defect detection, significant challenges persist in adapting these techniques to PCB surface inspection scenarios. Existing methods are effective in detecting medium to large defects or in environments where object sizes and defect distributions are relatively regular. However, the characteristics of modern PCBs—such as increasing miniaturization of components, denser circuit layouts, and complex multilayer structures—pose substantial challenges [21], [22]. Defects become smaller, more irregular, and densely distributed as PCBs evolve toward greater integration and higher precision [23]. These trends significantly increase the difficulty for conventional models to accurately locate and classify small defects, as they often lose fine-grained features during feature extraction and struggle to distinguish subtle defect patterns from complex backgrounds [24]. Therefore, the enhancement of small object detection capability has become critical for achieving accurate and reliable PCB defect inspection in intelligent manufacturing environments.

This study presents a novel framework, denoted as ABF-YOLO, to achieve precise and robust defect recognition in complex industrial environments, thereby addressing small-defect detection difficulties and mitigating background noise during a PCB surface analysis. Experiments were performed on benchmark datasets to validate the detection performance of the proposed approach for small and densely clustered defect types.

The main contributions of this study are threefold. First, a new PCB surface defect detection framework with integration of axial attention and bidirectional feature fusion (BiFusion) modules was tailored to enhance the recognition accuracy of fine-grained defects within noisy environments. Second, the axial attention mechanism was introduced to capture directional long-range spatial relationships by independently modeling horizontal and vertical contexts, which can maintain low computational overhead and preserve structural granularity, thus improving the performance of subtle defect detection. Third, the BiFusion module was designed to leverage learnable weighting and channel-wise fusion to highlight meaningful features and selectively filter out irrelevant ones, facilitating multi-level feature aggregation from the top-down and bottom-up directions.

The rest of this paper is organized as follows. Section II presents the literature review on object detection, attention mechanisms, and multi-scale feature fusion strategies. Section III introduces the ABF-YOLO framework with incorporation of axial attention and the BiFusion modules. Section IV provides the experimental results of the proposed approach, followed by the conclusions in Section V.

II. Related Work

This section provides a comprehensive literature review of PCB surface defect detection using deep learning-based approaches, including traditional deep learning, deep learning-based network with attention mechanism, and multi-scale feature fusion strategies.

A. Traditional Deep Learning-Based Object Detection

In recent years, CNN-based approaches have been extensively utilized for object detection tasks, particularly in PCB defect identification. Faster R-CNN [12] operates by initially generating region proposals, followed by classification and localization, offering high precision at the expense of computational efficiency and inference latency. One-stage detectors, exemplified by the YOLO family, utilize an end-to-end pipeline that directly predicts object categories and bounding boxes, significantly accelerating inference speed. YOLOv5 has been commonly adopted in industrial defect detection, and variants, such as YOLOv5-PCB, have enhanced its ability to detect small defects through deep and light backbone designs [25]. YOLOv8 further integrates re-parameterization modules and feature pyramid networks (FPNs), striking an optimal balance between accuracy and efficiency [26].

Nevertheless, these conventional detectors still struggle with fine-grained PCB defect detection. Small defects are typically diluted in deep layers, and key features are overwhelmed by background interference, resulting in increased false positives and missed detections [27]. Current models still lack sufficient discriminatory power, particularly for small and irregular defects, such as mouse bites or spurious copper.

PCB defect datasets often exhibit significant class imbalance, where frequent categories, such as “missing hole”, dominate the sample distribution, leading to biased prediction. One-stage detectors, including YOLOv5 and YOLOv8, have incorporated focal loss into the classification head to mitigate this issue, minimizing the influence of dominant easy negatives and enhancing the learning of rare and hard samples and detection robustness [28].

B. Deep Learning-Based Object Detection With Attention Mechanism

Recently, attention mechanisms have proven highly effective in computer vision tasks, boosting detection accuracy by guiding models to concentrate on key features and

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filter out distractions. Attention mechanisms are classified into three main types: channel-based, spatial-based, and hybrid forms. Channel-wise attention enhances feature encoding by capturing inter-channel relationships, as exemplified by squeeze-and-excitation networks (SENets) [29]. Spatial attention techniques emphasize critical spatial areas in feature maps, boosting localization effectiveness. The convolutional block attention module (CBAM) applies channel and spatial attention in sequence, resulting in refined and informative feature extraction [30]. The introduction of Transformer-based architectures has further advanced attention mechanisms by allowing models to capture global dependencies through self-attention, as demonstrated in the DETR model [31]. The advantages and disadvantages of these three attention mechanisms are evident. SENet captures global channel dependencies but lacks spatial discrimination. CBAM enhances channel and spatial saliency but introduces sequential processing overhead. DETR with full self-attention excels at modeling global dependencies yet suffers from slow convergence and requires large-scale datasets.

However, existing attention mechanism-based approaches are still limited to fine-grained and small-scale defect detection, particularly in complex backgrounds [32]. These reported methods often struggle to generalize across different feature hierarchies and/or defect types. Axial attention mechanism is introduced in this study to address these limitations. Unlike the full 2D attention in DETR, the axial attention module applies 1D attention along spatial axes separately, thereby minimizing computational complexity while retaining long-range dependency modeling, which is particularly suitable for detecting complex defects, such as thin open circuits and elongated traces.

C. Multi-Scale Feature Fusion

In object detection, multi-scale fusion strategies are commonly applied to boost detection effectiveness, particularly for size-diverse targets. The FPN integrates top-down flows and lateral links to merge semantic-level features with fine spatial cues, enhancing detection across multiple object scales [33]. However, the unidirectional information flow of FPN constrains its fusion capability, potentially resulting in suboptimal feature integration. The path aggregation network (PANet) introduces a complementary bottom-up pathway atop the FPN structure, thereby enhancing information flow across feature hierarchies and small object detection and addressing the shortcomings of FPN [34]. Although PANet facilitates bidirectional fusion, it relies on static weighting across feature scales, limiting its adaptability to diverse detection scenarios. The adaptively spatial feature fusion technique was introduced to promote dynamic feature fusion by learning spatial weights across feature scales, thereby reducing inter-scale conflicts and enhancing robustness to scale variations [35]. Nevertheless, the increased computational cost of this technique may hinder real-time applications.

Furthermore, RepGFPN was developed to achieve efficient and effective multi-scale feature fusion, which utilizes a learnable weighting mechanism to adaptively fuse features of different scales in a bidirectional path [36]. Although RepGFPN has effectively performed in various object detection tasks, it still faces challenges in handling small targets and complex backgrounds.

In summary, current multi-scale fusion techniques have already achieved notable advancements in object detection. However, these techniques are still limited to the detection of small defects within background interference scenarios.

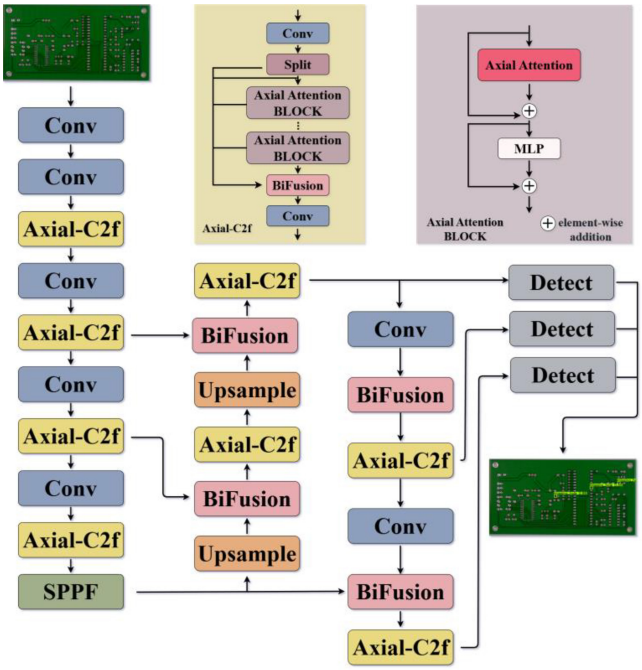


Fig. 1. Overall architecture of the proposed ABF-YOLO network.

III. ABF-YOLO Framework

A. Overall Architecture

ABF-YOLO is designed as a three-stage architecture consisting of a Backbone for feature extraction, a Neck for multi-scale fusion, and a Head for generating the final predictions (Fig. 1).

The Backbone incorporates an innovative axial attention mechanism that models long-range spatial correlations while maintaining computational efficiency to address the difficulty of identifying small defects and subtle structural cues. Meanwhile, the Neck utilizes a BiFusion strategy, adaptively aggregating multi-scale features to enhance semantic understanding and detailed localization. These enhancements strengthen the model's capacity for accurate defect localization and classification under complex industrial scenarios.

B. Axial Attention Mechanism

This subsection elaborates on the architectural design of the axial attention mechanism. The axial attention block implemented in the network is a compound unit that combines an axial attention module with a lightweight MLP layer (Fig. 1). The axial attention module is specifically designed to model extended spatial correlations along horizontal and vertical

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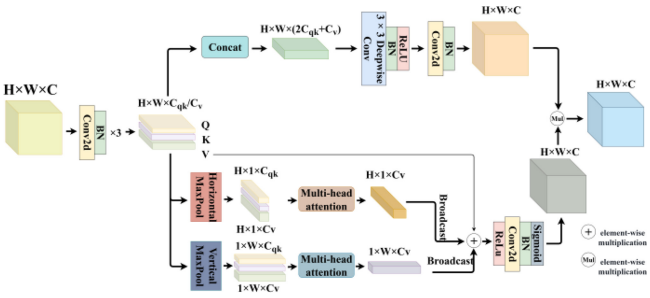


Fig. 2. Diagram of the proposed axial attention module.

axes. Meanwhile, the following MLP layer enhances feature encoding via nonlinear transformations. Unlike the traditional 2D self-attention module, the proposed axial attention module reduces complexity by decomposing 2D spatial modeling into two successive 1D operations, significantly lowering the computational cost while preserving global modeling capacity. Moreover, the combination of squeeze axial attention and the detail enhancement branch enables the model to maintain a fine-grained spatial structure and enhance contextual awareness.

Fig. 2 presents the architecture and processing pipeline of the proposed axial attention module. The module comprises two essential parts to effectively model long-range interactions while retaining detailed representations: squeeze axial attention and a detail enhancement Kernel [37].

Starting with an input feature map $X \in \mathbb{R}^{H\times W\times C}$, the data are initially split into three projection branches, namely, query (Q), key (K), and value (V), using 1×1 convolutions, yielding

$$i = W_i X \quad (i = Q, K, V), \tag{1}$$

where i represents the individual components of Q, K, and V, with W_i being the corresponding weight matrix for each component.

The squeeze axial attention module performs global average pooling independently along the horizontal and vertical axes to condense feature representations, rather than conventional 2D self-attention

$$\begin{cases} i^{(h)} = \text{AvgPool}_H(i) \\ i^{(v)} = \text{AvgPool}_V(i), \end{cases} \tag{2}$$

where AvgPool_H and AvgPool_V denote global average pooling along the two dimensions, and $i = Q, K, V$.

After obtaining the squeezed features, the horizontal and vertical attention maps are separately computed using multi-head attention operations:

$$\begin{cases} \text{Attention}^{(h)} = \text{Softmax}\left(\dfrac{Q^{(h)}\left(K^{(h)}\right)^T}{\sqrt{d}}\right)V^{(h)} \\ \text{Attention}^{(v)} = \text{Softmax}\left(\dfrac{Q^{(v)}\left(K^{(v)}\right)^T}{\sqrt{d}}\right)V^{(v)}, \end{cases} \tag{3}$$

where d denotes the dimension of each attention head.

The results from the horizontal and vertical attention are upsampled to align with the input spatial size and integrated with the original V feature map via element-wise summation:

$$V' = V + Broadcast\left(Attention^{(h)}\right) + Broadcast\left(Attention^{(v)}\right). \tag{4}$$

The detail enhancement branch applies a 3×3 depth-wise separable convolution to the concatenated (Q, K, V) features, producing an enhanced representation map:

$$E = \text{DWConv}_{3\times 3}\left(\text{Concat}^{(Q,K,V)}\right). \tag{5}$$

Finally, the outputs of the attention path V' and the detail enhancement path E are fused by element-wise multiplication

$$Y = V' \otimes E. \tag{6}$$

The outputs from squeeze axial attention and detail enhancement are fused by element-wise multiplication to achieve a balanced global–local feature representation.

The proposed axial attention mechanism significantly reduces computational complexity compared with traditional 2D self-attention and other lightweight attention mechanisms. The axial attention module decomposes the spatial modeling into two 1D attention operations, reducing complexity from $O(HW\times HW)$ to $O(H\times W(H+W))$, where H and W denote the height and width of feature map, respectively. This mechanism maintains a strong global modeling capability and enables lightweight deployment, offering substantial benefits for real-time defect inspection of PCB images with high-resolution feature maps.

C. BiFusion

The proposed BiFusion module is designed to improve multi-scale feature integration via adaptive learning strategies [38]. Inspired by the bidirectional cross-scale connections of BiFPN, BiFusion incorporates learnable weights to adaptively balance the feature contributions across multiple resolutions and performs iterative fusion in the top-down and bottom-up directions, mitigating unidirectional constraints and promoting semantic-detail synergy across layers. Additionally, BiFusion generates outputs by concatenating multi-scale features along the channel axis to align with the network structure, enriching channel diversity and aiding the downstream detection process.

1) *Bidirectional Cross-scale Connections*
Top-down path: High-level semantic features are progressively up-sampled and fused with lower-level features, transferring semantic information to higher-resolution layers.
Bottom-up path: Lower-level features, after initial fusion, propagate upward to provide a detailed context to higher-level semantic features, forming a closed loop for iterative refinement.
2) *Learnable Weighted Fusion*
BiFusion adopts adaptive weight coefficients to emphasize essential features during bidirectional multi-scale fusion. Specifically, BiFusion is formulated as follows:

$$F_{\text{fasion}} = \frac{w_1 \cdot F_1 \oplus w_2 \cdot F_2 \oplus w_3 \cdot F_3}{w_1 + w_2 + w_3 + \varepsilon}, \tag{7}$$

where $w_i (i = 1, 2, 3)$ denotes the weight coefficient for the ith input feature, and ε is a minor constant ensuring numerical stability. The operator $\oplus$ denotes channel-wise concatenation.

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Each input feature map (F_1, F_2, and F_3) shares an identical spatial resolution $H \times W$ while possessing distinct channel dimensions (C_1, C_2, and C_3). The BiFusion module concatenates these three feature maps along the channel axis to effectively integrate multi-scale information, yielding a unified tensor of shape $H \times W \times (C_1 + C_2 + C_3)$ and followed by the weighted fusion strategy.

3) *Multi-scale Detection*

Small object detection: After enriching low-level detailed features with higher-level semantic information, localization and classification of small objects become highly accurate.

Large object detection: High-level features, enhanced by spatial details from the bottom-up fusion path, effectively delineate object boundaries. **Overall trainability**: The adaptive weight coefficients are jointly trained with the network parameters, allowing the network to adaptively learn to distribute feature importance across scales, ultimately achieving enhanced accuracy and robustness.

IV. Experiments and Discussions

A. Experimental Environment Setup

The experiments were conducted on the system equipped with an NVIDIA RTX 4090D GPU (24 GB VRAM) and an AMD EPYC 9754 128-core CPU, with 18 virtual cores allocated. All models were implemented and evaluated using PyTorch 2.1.2, Python 3.10, and CUDA 11.8 and trained for 300 epochs with a fixed batch size of four. The input images were uniformly resized to 640 × 640 pixels. The original PCB images (3034 × 1586) were resized to 640 × 640 for efficiency. The network utilizes axial attention and BiFusion modules to mitigate potential fine-grained detail loss, enhancing feature preservation during downsampling.

B. Dataset and Implementation Details

The public PKU-Market-PCB dataset, which was curated by the Intelligent Robot Open Laboratory at Peking University, was utilized in the experiments [39]. The dataset comprises 1386 industrial-grade PCB samples spanning multiple defect types, with a balanced inter-class distribution ($\sigma = 12.7$) that effectively minimizes the effect of long-tailed data during training. The dataset was divided strictly according to the official standard into training (831 images), validation (277 images), and testing sets (278 images). The defects in this dataset exhibited specific characteristics with small size and dense spatial distribution. Specifically, the defects typically occupied only a small portion of each image, and background noise closely resembled defect features. Moreover, defects frequently appeared with a median occurrence of three defects per image, highlighting dense target distribution and partial occlusion issues.

An analysis of each defect type is provided below to further elucidate these characteristics. The dataset defines six common defect categories, each with unique visual attributes and spatial complexities [40], [41]. The missing hole defects are large, circular, and geometrically distinct, occupying well-defined regions that contrast with the surrounding background. The mouse bite defects appear as clusters of small, irregular perforations along the trace edges. The shapes of these defects are inconsistent, and their boundaries are often blurred, which may cause visual confusion with the surrounding structures. Spurious copper is typically represented by amorphous or fragmented copper patches that deviate from the intended circuit layout. These regions vary in size and texture and are often indistinguishable from residual copper or background artifacts. Open circuit defects denote discontinuity and/or break in conductive paths. The linear and fragmented nature of the aforementioned defects makes them less visually prominent, especially when embedded in dense trace regions. Spur defects are fine protrusions or sharp artifacts extending from traces or pads. These spur defects are often visually subtle due to their narrow structure and close alignment with background patterns. Short defects are unintended connections between nearby conductors and typically occur in densely routed areas, where their compactness and proximity to legitimate traces increase their visual complexity. These structural characteristics collectively contribute to the difficulty of PCB defect detection—particularly in scenarios involving small target size, low foreground-background separability, and high spatial density—and highlight the need for fine-grained representation modeling.

The original annotations in the dataset were provided in XML files following the PASCAL VOC format, including defect categories, bounding box coordinates, and image metadata. The original dataset was re-divided using a Python script, strictly following a stratified sampling strategy based on defect categories, to prevent potential overfitting due to uneven data partitioning.

C. Evaluation Metrics

The widely adopted evaluation indicators for object detection tasks comprise precision, recall, average precision (AP), and mean average precision (mAP). mAP at an IoU threshold of 0.5 is commonly denoted as mAP@0.5 and computed as follows:

$$mAP@0.5 = \frac{1}{N}\sum_{i=1}^{N} AP_i, \tag{8}$$

where N represents the total number of object categories; and AP_i represents the average precision for the ith category, obtained by calculating the area beneath the corresponding P–R curve at an IoU threshold of 0.5 or above.

mAP@0.5, a widely accepted metric in object detection, offers a holistic assessment of model performance by jointly evaluating the classification and localization accuracy. The score ranges from zero to one, with higher mAP@0.5 indicating improved identification and localization of true positives while reducing false alarms and mislocalization.

D. Performance Comparisons

The performance of the proposed ABF-YOLO model was first compared with several representative models from the YOLO family. Fig. 3 and Table I shows the AP results

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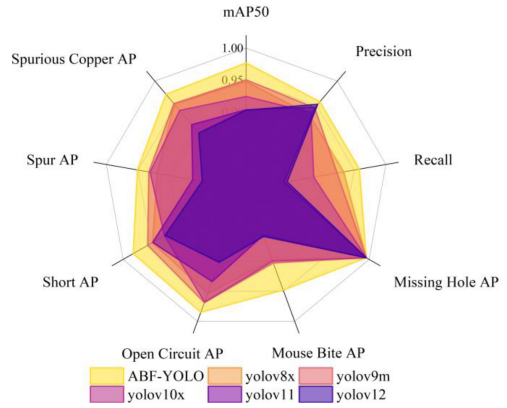


Fig. 3. Radar chart of the defect detection metrics.

TABLE I
CATEGORY-WISE AP COMPARISON OF ABF-YOLO WITH OTHER YOLO-SERIES MODELS ON SIX COMMON PCB DEFECTS

Model	Missing hole	Mouse bite	Open circuit	Short	Spur	Spurious copper
Yolov8x	0.995	0.898	0.954	0.947	0.92	0.953
Yolov9m	0.995	0.903	0.969	0.956	0.933	0.954
Yolov10x	0.995	0.9	0.968	0.926	0.931	0.94
Yolov11	0.995	0.857	0.934	0.947	0.864	0.911
Yolov12	0.995	0.859	0.902	0.925	0.849	0.894
ABF-YOLO	0.995	**0.948**	**0.985**	**0.982**	**0.951**	**0.973**

TABLE II
OVERALL DETECTION PERFORMANCE COMPARISON AMONG ABF-YOLO AND YOLO VARIANTS

Model	mAP@0.5	Precision	Recall
Yolov8x	0.9479	0.9318	0.9335
Yolov9m	0.9503	0.9454	0.9231
Yolov10x	0.9244	0.9360	0.8865
Yolov11	0.9036	0.9466	0.8399
Yolov12	0.9030	0.9525	0.8444
ABF-YOLO	**0.9767**	**0.9578**	**0.9583**

across six types of PCB defects. All models exhibit similar AP performance on the missing hole defect (AP = 0.995), indicating that large, high-contrast defects are consistently detectable. However, the performance significantly varies in categories with smaller or irregular features. ABF-YOLO achieves the highest AP scores for the types of mouse bite, open circuit, short, spur, and spurious copper, demonstrating its superior capability in handling fine-grained and fragmented defects. By contrast, YOLOv12 and YOLOv11 show a noticeable decline in these categories, especially for "spur" and "spurious copper", which are more susceptible to background noise and shape ambiguity.

Table II presents the overall detection performance. ABF-YOLO achieves the highest mAP@0.5 (97.67%), precision (95.78%), and recall (95.83%) among all models. ABF-YOLO improves mAP by +7.37%, precision by +0.53%, and recall by +11.39% compared with YOLOv12. Additionally, ABF-YOLO shows a 2.6%–7.2% advantage in mAP and recall when compared with YOLOv9m and YOLOv10x, indicating strong robustness in handling small-scale and densely distributed defects under complex PCB inspection conditions.

TABLE III
OVERALL DETECTION PERFORMANCE COMPARISON AMONG CLASSICAL DETECTORS AND ABF-YOLO

Model	mAP@0.5	Precision	Recall
DETR	0.891	0.899	0.887
Faster-RCNN	0.713	0.738	0.631
Yolov5	0.950	0.932	0.935
ABF-YOLO	**0.977**	**0.958**	**0.958**

TABLE IV
CATEGORY-WISE AP COMPARISON AMONG CLASSICAL DETECTORS AND ABF-YOLO

Model	Missing Hole	Mouse Bite	Open Circuit	Short	Spur	Spurious Copper
DETR	0.981	0.910	0.879	0.904	0.872	0.799
Faster-RCNN	0.881	0.650	0.651	0.730	0.707	0.657
Yolov5	0.995	0.9	0.965	0.942	0.929	0.947
ABF-YOLO	**0.995**	**0.948**	**0.985**	**0.982**	**0.951**	**0.973**

The qualitative results (Fig. 4) support these findings. ABF-YOLO provides more accurate localization and delineation of defect regions, with fewer false positives and clearer contour definitions compared with the baseline models. The model also exhibits higher sensitivity in detecting minute defects that are often overlooked. The precision–recall (PR) curve on the test set is demonstrated in Fig. 5 to further validate the sensitivity and robustness of ABF-YOLO across varying classification thresholds. The figure shows a consistently high PR trade-off, with strong confidence calibration and effective discrimination across a wide range of thresholds.

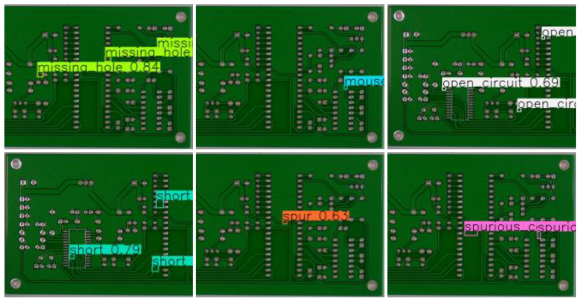
Comparative experiments with three classical object detectors, namely, DETR, Faster-RCNN, and YOLOv5, were performed to further verify the generalizability of ABF-YOLO beyond the YOLO family. ABF-YOLO improves mAP by +2.7%, precision by +2.6%, and recall by +2.3% compared with the YOLOv5 model (Table III). The performance improvements of ABF-YOLO are more pronounced compared with DETR and Faster R-CNN, and ABF-YOLO leads in all metrics by large margins.

Table IV lists the comparative results of the category-wise AP between ABF-YOLO and classical detectors. ABF-YOLO achieves the best performance across all six defect types. For instance, ABF-YOLO attains 0.985 and 0.982 in detecting open circuit and short defects and records 0.951 and 0.973 in detecting more ambiguous defects, such as spur and spurious copper, showing noticeable improvements over DETR (0.872 and 0.799) and Faster-RCNN (0.707 and 0.657). All models effectively perform in detecting missing hole defect, while ABF-YOLO still matches the highest AP with 0.995.

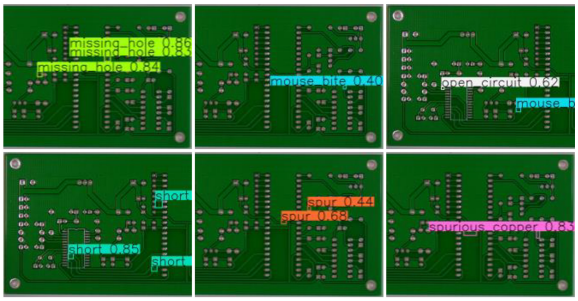
E. Ablation Experiments

The ablation experiments were conducted to validate the individual contributions of each module to the overall performance improvement of ABF-YOLO. The incorporation of only the axial attention module resulted in a notable

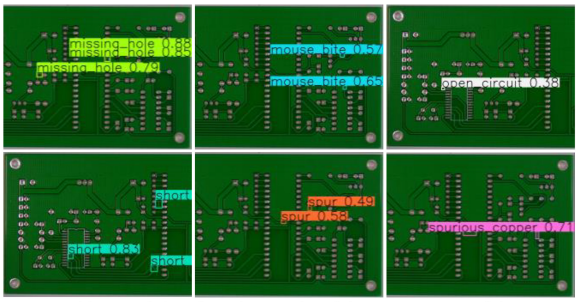
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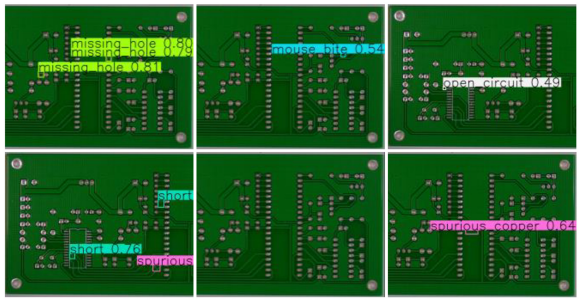
(a) PCB defect detection results of the YOLOv8 model: missing hole, mouse bite, open circuit, short, spur, and spurious copper (from left to right and top to bottom, the defect orders of the following subfigures are the same)



(b) PCB defect detection results of the YOLOv9 model



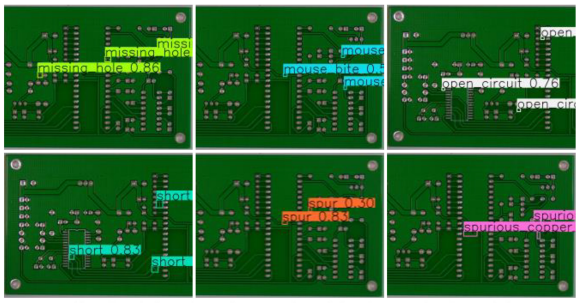
(c) PCB defect detection results of the YOLOv10 model



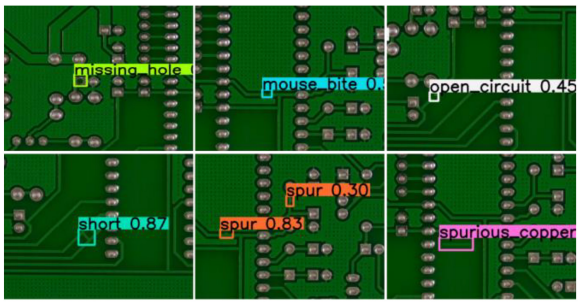
(d) PCB defect detection results of the YOLOv11 model



(e) PCB defect detection results of the YOLOv12 model



(f) PCB defect detection results of the ABF-YOLO model



(g) Zoom-in views of ABF-YOLO's predictions of six types of defects

Fig. 4. Qualitative comparison results of PCB defect detection using different models. Each subfigure contains six defect categories, including missing hole, mouse bite, open circuit, short, spur, and spurious copper (from left to right and top to bottom), corresponding to the following board IDs: 01_missing_hole_03, 01_mouse_bite_04, 01_open_circuit_03, 01_short_07, 01_spur_17, and 01_spurious_copper_07, respectively.

mAP@0.5 increasement from 94.79% to 96.41%, with consistent gains in multiple categories, particularly for mouse bite defect detection (from 89.8% to 94.8%), benefiting from improved spatial sensitivity and directional attention modeling (Tables V and VI). The integration of the BiFusion module resulted in an mAP@0.5 increasement from 94.79% to 95.03%, and the improvements are evident in classes, such as spur and spurious copper, indicating its effectiveness in aggregating fine-grained multi-scale features.

The combination of both modules yielded the highest performance, with mAP@0.5 reaching 97.67%, alongside substantial increases in precision (95.78%) and recall (95.83%). The results summarized in Table VI demonstrate that the integration of the axial attention and BiFusion modules significantly enhances the detection performance of the framework, particularly for identifying fine-grained and structurally complex defects in complex industrial scenarios.

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TABLE V
ABLATION STUDY ON DEFECT-TYPE-SPECIFIC DETECTION PERFORMANCE

Model	Missing hole	Mouse bite	Open circuit	Short	Spur	Spurious copper
Yolov8x	0.995	0.898	0.954	0.947	0.92	0.953
Yolov8–axial attention	0.995	0.916	0.983	0.966	0.943	0.976
Yolov8–BiFusion	0.995	0.891	0.966	0.952	0.943	0.955
ABF-YOLO	0.995	**0.948**	**0.985**	**0.982**	**0.951**	0.973

TABLE VI
OVERALL PERFORMANCE COMPARISON IN ABLATION EXPERIMENTS

Model	mAP@0.5	Precision	Recall
Yolov8x	0.9479	0.9319	0.9335
Yolov8–axial attention	0.9641	0.9487	0.9328
Yolov8–BiFusion	0.9503	0.9548	0.9146
ABF-YOLO	**0.9767**	**0.9578**	**0.9583**

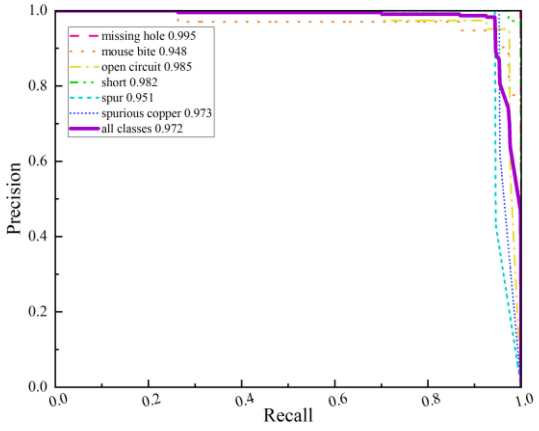


Fig. 5. P–R curve of ABF-YOLO on the test dataset.

F. Discussions

The proposed ABF-YOLO framework demonstrates consistent superiority across category-wise and overall performance metrics, particularly excelling in the detection of subtle and fragmented defect types, such as "mouse bite", "spurious copper", and "spur". These categories typically suffer from ambiguous boundaries and indistinct features, posing substantial challenges to conventional models.

This performance enhancement is primarily attributed to two architectural innovations: axial attention module and BiFusion mechanism. The axial attention module enhances the model's ability to capture long-range spatial dependencies along horizontal and vertical directions, enabling it to effectively recognize elongated and structurally disconnected defects, such as "open circuit" and "short". Meanwhile, the BiFusion module provides adaptive multi-scale feature integration through learnable weights and bidirectional information flow, ensuring a balance between semantic abstraction and fine-grained detail preservation.

Overall, the integration of directional attention and adaptive feature fusion results in a discriminative and robust representation, enabling ABF-YOLO to effectively address the challenges of fine-grained, small-scale, and spatially complex PCB defect detection in real-world industrial environments.

The process of tuning hyperparameters to validate the module design, such as loss weight balancing and learning rate scheduling, is time-consuming and required extensive empirical trials. These difficulties are addressed through iterative ablation experiments and careful training configuration adjustments, thereby realizing a robust and generalizable detection framework.

V. CONCLUSION

This study presents an enhanced YOLO framework, denoted as ABF-YOLO, which integrates the axial attention mechanism and BiFusion strategy, to address the low accuracy challenges in detecting small-scale defects on PCB surfaces using the existing mainstream YOLO baseline models. The axial attention module improves the capturing of fine-grained defect features by modeling directional long-range dependencies. Meanwhile, the BiFusion strategy adaptively aggregates multi-scale features to enhance the semantic representation and localization precision. The extensive experiments on the PKU-Market-PCB dataset demonstrate that the proposed ABF-YOLO model achieves superior detection accuracy, especially for small, ambiguous, and structurally complex defects. Future work will focus on extending the framework to more diverse industrial datasets, incorporating domain adaptation techniques to enhance generalizability and improving the real-time performance for deployment on resource-constrained hardware.

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